PROBLEM 17

Linear Central Force under Gravity

Original statement · Problems.pdf, p. 9

Central Force + Gravity

A force attracts a heavy particle M of mass m to a point O. The attractive force is proportional to the mass of the particle and to the distance OM. The proportionality coefficient is k^2. The coordinate system $Oxyz$ has the origin at the centre of attraction O and the axis Oz is vertically oriented. Initially, the coordinates of M are

$$(x_0, y_0, z_0) = \left(\frac{\sqrt{3}g}{k^2}, 0, 0\right)$$

and its initial velocity is

$$(\dot{x}_0, \dot{y}_0, \dot{z}_0) = \left(0, \frac{2g}{k}, 0\right)$$

(a) Describe the motion of M, specifying clearly the shape of the path.

(b) What is the velocity of the particle as a function of t?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a heavy particle M of mass m under (i) a linear central attraction $\mathbf{F}_a = -mk^2\mathbf{r}$ towards O and (ii) its weight $\mathbf{F}_g = -mg\mathbf{k}$ (Oz points up).
- **Hypotheses:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is inertial; no drag or friction; m, g, k are positive constants.
- **Degrees of freedom:** free particle in 3D space, $\mathrm{CDOF} = 3$ with coordinates (x, y, z).
- **Initial conditions ($t = 0$):** $\mathbf{r}(0) = \dfrac{\sqrt{3}g}{k^2}\mathbf{i}$ and $\mathbf{v}(0) = \dfrac{2g}{k}\mathbf{j}$.

Symbol	Meaning	SI unit
m	mass of M	kg
k	attraction coefficient is k^2	s^{-1}
g	gravitational acceleration	$\mathrm{m/s}^2$
g/k^2	length scale of the problem	m
g/k	speed scale of the problem	m/s

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathcal{B}_0\}$, $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$; $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}$.
- **Orbit-plane basis** (introduced after the planar result of Phase 3): $\mathcal{B}_1 = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ with

$$\mathbf{e}_1 = \left(\tfrac{\sqrt{3}}{2}, 0, \tfrac{1}{2}\right), \qquad \mathbf{e}_2 = \mathbf{j}, \qquad \mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2 = \left(-\tfrac{1}{2}, 0, \tfrac{\sqrt{3}}{2}\right)$$

Change-of-basis matrix (columns are the components of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \sqrt{3}/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & \sqrt{3}/2 \end{pmatrix}$$

Checks. $\mathbf{e}_1 \cdot \mathbf{e}_1 = \frac{3}{4} + \frac{1}{4} = 1$, $\mathbf{e}_3 \cdot \mathbf{e}_3 = \frac{1}{4} + \frac{3}{4} = 1$, $\mathbf{e}_1 \cdot \mathbf{e}_3 = -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} = 0$, hence $[{}_0R_1]^T[{}_0R_1] = I$; and $\det[{}_0R_1] = \frac{3}{4} + \frac{1}{4} = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

1. Equations of motion

Pedagogical Justification: $m\mathbf{a} = \mathbf{F}_a + \mathbf{F}_g$ in the inertial frame. Because the restoring force is linear in $\mathbf{r}$, each Cartesian component decouples.

$$m\left(\ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k}\right) = -mk^2\left(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}\right) - mg\mathbf{k} \implies \ddot{x} + k^2x = 0, \quad \ddot{y} + k^2y = 0, \quad \ddot{z} + k^2z = -g$$

2. Integration of $x(t)$ and $y(t)$

Pedagogical Justification: $\ddot{q} + k^2q = 0$ is linear with constant coefficients; its characteristic roots are $\lambda = \pm ik$.

$$x(t) = A_1\cos kt + B_1\sin kt, \qquad \dot{x}(t) = -kA_1\sin kt + kB_1\cos kt$$

Initial data $x(0) = \sqrt{3}g/k^2, \dot{x}(0) = 0$ give $A_1 = \sqrt{3}g/k^2$ and $kB_1 = 0$. Likewise $y(0) = 0 \Rightarrow A_2 = 0$ and $\dot{y}(0) = 2g/k \Rightarrow kB_2 = 2g/k \Rightarrow B_2 = 2g/k^2$:

$$x(t) = \frac{\sqrt{3}g}{k^2}\cos kt, \qquad y(t) = \frac{2g}{k^2}\sin kt$$

3. Integration of $z(t)$

> **Pedagogical Justification:** The equation $\ddot{z} + k^2 z = -g$ is forced by a constant. A constant particular solution z_p satisfies $k^2 z_p = -g$, i.e. $z_p = -g/k^2$ (the equilibrium height), and the homogeneous part is as before.

$$z(t) = -\frac{g}{k^2} + A_3 \cos kt + B_3 \sin kt, \qquad \dot{z}(t) = -kA_3 \sin kt + kB_3 \cos kt$$

$z(0) = 0 \Rightarrow A_3 = g/k^2$ and $\dot{z}(0) = 0 \Rightarrow B_3 = 0$:

$$z(t) = \frac{g}{k^2}\left(\cos kt - 1\right)$$

4. Shape of the path

Planar confinement. x and z share the phase function $\cos kt$: from $\cos kt = k^2 x/(\sqrt{3}g)$,

$$z = \frac{g}{k^2}\left(\frac{k^2 x}{\sqrt{3}g} - 1\right) = \frac{x}{\sqrt{3}} - \frac{g}{k^2} \implies x - \sqrt{3}\, z - \frac{\sqrt{3}g}{k^2} = 0$$

The plane has normal $\mathbf{n} = (1, 0, -\sqrt{3})/2$ and contains the direction $(\sqrt{3}, 0, 1)$, which makes $\arctan(1/\sqrt{3}) = 30^\circ$ with the horizontal: the plane is inclined 30° to the horizontal.

Circle. With $\mathbf{C} = (0, 0, -g/k^2)$, the distance to $\mathbf{C}$ is constant:

$$x^2 + y^2 + \left(z + \frac{g}{k^2}\right)^2 = \frac{g^2}{k^4}\left[3\cos^2 kt + 4\sin^2 kt + \cos^2 kt\right] = \frac{4g^2}{k^4}$$

The distance from $\mathbf{C}$ to the plane is $\left|0 - \sqrt{3}(-g/k^2) - \sqrt{3}g/k^2\right| / \sqrt{4} = 0$, so the plane passes through the centre of the sphere and the path is a **great circle of radius** $R_c = 2g/k^2$, covered at angular rate k. In the plane basis of Phase 2 the three solutions combine into one vector identity:

$$\mathbf{r}(t) - \mathbf{C} = R_c\,[\cos kt\,\mathbf{e}_1 + \sin kt\,\mathbf{e}_2]$$

since $R_c \cos kt\,\mathbf{e}_1 = \left(\frac{\sqrt{3}g}{k^2}\cos kt,\ 0,\ \frac{g}{k^2}\cos kt\right)$ reproduces $x(t)$ and $z(t) + g/k^2$, and $R_c \sin kt\,\mathbf{e}_2$ reproduces $y(t)$. The $\mathbf{e}_3$-component vanishes (planar motion) and $\mathbf{e}_3 \cdot \mathbf{k} = \sqrt{3}/2$ confirms the 30° inclination of the normal to the vertical. Only the projection on Oxy, $\left(\dfrac{x}{\sqrt{3}g/k^2}\right)^2 + \left(\dfrac{y}{2g/k^2}\right)^2 = 1$, is an ellipse; the circle is seen obliquely.

5. Velocity (part b)

Pedagogical Justification: Differentiate each component with respect to time (chain rule $\frac{d}{dt}\cos kt = -k\sin kt$).

$$\mathbf{v}(t) = -\frac{\sqrt{3}g}{k}\sin kt\,\mathbf{i} + \frac{2g}{k}\cos kt\,\mathbf{j} - \frac{g}{k}\sin kt\,\mathbf{k}$$

$$v^2 = \frac{g^2}{k^2}\left[3\sin^2 kt + 4\cos^2 kt + \sin^2 kt\right] = \frac{4g^2}{k^2}\left(\sin^2 kt + \cos^2 kt\right) = \frac{4g^2}{k^2} \implies v = \frac{2g}{k} = \text{const}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Why the speed is constant.** The force is conservative with potential $V = \frac{1}{2}mk^2(x^2 + y^2 + z^2) + mgz = \frac{1}{2}mk^2\left[x^2 + y^2 + (z + g/k^2)^2\right] - \dfrac{mg^2}{2k^2}$, which depends only on the distance to $\mathbf{C}$. That distance is constant on the circle, so V and therefore the kinetic energy are constant; also $v = kR_c = 2g/k$.
- **Initial-condition check:** $\mathbf{r}(0) = (\sqrt{3}g/k^2, 0, 0)$ and $\mathbf{v}(0) = (0, 2g/k, 0)$ are reproduced by the formulas above.
- **Limit $g \to 0$:** the centre $\mathbf{C} \to O$ and the circle lies in a plane through O, but the radius $R_c = 2g/k^2 \to 0$ as well (the initial data $\sqrt{3}g/k^2$ and $2g/k$ also vanish), so the path shrinks to the point O; it is not an elliptic oscillator orbit of finite size.

- **Dimensions:** $[g/k^2] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}}} = \mathrm{m}$, $[g/k] = \mathrm{m/s}$, $[kt] = 1$.
- **Numerical check (Python, $g = 9.81$, $k = 2$):** over 20 s, $\max \left| \|\mathbf{r} - \mathbf{C}\| - R_c \right| = 9 \times 10^{-16}$ m, $\max |v - 2g/k| = 2 \times 10^{-15}$ m/s, the plane residual $x - \sqrt{3}z - \sqrt{3}g/k^2$ stays below 3×10^{-15} m, and the finite-difference residual of $\ddot{x} + k^2 x = 0$ and $\ddot{z} + k^2 z + g = 0$ is below 4×10^{-7}.
- **Official key** (Problems.pdf, PDF page 73): circle given by the sphere $x^2 + y^2 + (z + g/k^2)^2 = 4g^2/k^4$ and the plane $x - \sqrt{3}z = \sqrt{3}g/k^2$, constant speed $2g/k$.

Source: Problems.pdf (PDF pages 9-10); key on PDF page 73